



University of Vavuniya

Second Examination in Information Technology - 2021

First Semester – March 2023

IT2143 Visual Programming (Practical)

Answer All Questions

Time Allowed : Three hours

Instructions:

- All the necessary images are given in the folder *IT2143-Resources* on your PC's Desktop.
 - Create a new folder with your index number (e.g IT15099) on your PC's Desktop and use it to save your files.
 - You should use C# programming language and Visual Studio IDE to write the programs.
1. You are required to develop a console-based system that deals with some operations on course and student details. ;
- (a) The file *Course.cs* containing the class *Course* under the namespace *University* has the following:
- i. a *private string* variable for the title (Course Title)
 - ii. a *private int* variable for the duration (Course Duration)
 - iii. a *private string* variable for faculty (that offers the course)
 - iv. a *Constructor* for the *Course* class to initialize those variables
 - v. public properties for all variables
- (b) The file *Student1.cs* containing the class *Student* under the namespace *University* has the following:
- i. a *private string* variable for the registration number
 - ii. a *private string* variable for student name
 - iii. a *private int* variable for age
 - iv. a *private char* variable for sex

[This question is continued on the next page]

- v. a *private int* variable for the year of study
- vi. a *private string* variable for the filename of an image.
- vii. a *private Course* variable for course
- viii. a *Constructor* for the *Student* class to initialize those variables
- (c) The file *Student2.cs* containing the same class *Student* under the namespace *University* has the following:
- i. public properties for all variables
- (d) Generate a library file *University.dll* in which class *Course* and *Student* are bundled under the namespace *University*.
2. you are required to develop a windows-based system, which consists of several windows forms to manage the information of students in a University.
- (a) The following login form should be the initial user interface of the system.
- (b) The admin user can log in with username *admin* and password *admin@is*. And also the user can login with username *user* and password *user@is*.
- (c) The following windows form dashboard should be displayed after a successful login as an admin:

Form 3

Dashboard

Student Information System

view student

ADD COURSE

VIEW STUDENT

ADD STUDENT

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[This question is continued on the next page]

- (d) The following windows form dashboard should be displayed after a successful login as a user:



The screenshot shows a web browser window with a title bar that includes a 'Dashboard' tab and standard window controls (minimize, maximize, close). The main content area displays the text 'Student Information System' in a large, bold, black font. Below this text is a rectangular button with a black border and the text 'VIEW STUDENT' in bold, black, uppercase letters. There are several handwritten blue marks on the image: a small 'x' in the top left corner, a larger 'x' to the left of the button, and the word 'menu' written in cursive at the bottom right.

- (e) The button *ADD COURSE* in the dashboard should display the *Add Course* form shown below with the following functionalities:
- The fields for title, duration, and faculty.
 - The button *Back To Main* should display back the dashboard while hiding the form *ADD COURSE* itself.
 - The button *Add Course* should add or store the course information by displaying a message like *Course is added!* on a message box.
 - You should use the class *Course* in the library file *Unversity.dll*, to store the course information as an array of *Courses*.

ADD COURSE DETAILS

Title: Bsc in IT

Duration: 4

Faculty: Applied Science

Course is added!

OK

BACK TO MAIN ADD COURSE

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(f) The button *ADD STUDENT* in the dashboard should be displayed the *Add Student* form shown below with the following functionalities:

- Student's Photo should be loaded into the picture box with the help of the Load Image button to browse the photo.
- The other fields in the form should be included.
- The combo box for the Course should be loaded with the courses, which have been added using *Add Course* form.
- The button *Back To Main* should display back the dashboard while hiding the form *Add Student* itself.
- The button *ADD STUDENT* should add the student's information by displaying a message like *Student is added!* on a message box.
- You should use the class *Student* in the library file *Unversity.dll*, to store the student information as an array of Students.

Form 2

ADD STUDENTS DETAILS

ADD STUDENT DETAILS

Reg No: 2022/ICT/001

Name: K.Ramanayakke

Age: 23

Sex: ☒ Male ☐ Female

Year Of Study: 2

Course:
Bsc in IT
BSc in Physics

Photo: 

Load Image

BACK TO MAIN

ADD STUDENT

(g) The button *VIEW STUDENT* in the dashboard will display the *View Student* form shown below with the following functionalities:

- The combo box for the Registration Number should be loaded with the registration numbers of the students, which have been added using *Add Student* form.

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- ii. The details including course details should be displayed based on the selection of the registration number in the combo box for the Registration Number.
- iii. The button *Back To Main* should display back the dashboard while hiding the form *View Student* itself.



VIEW STUDENT DETAILS

Reg No:

2022/ICT/002

▼

2022/ICT/002

Name:

K.Ramanayekke

Age:

23

▼

Sex:

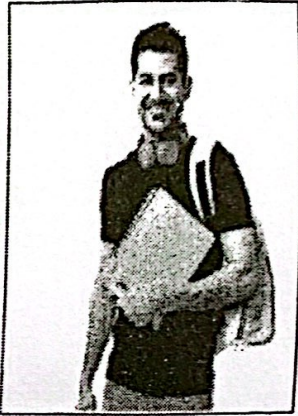
☒ Male ☐ Female

Year Of Study:

2

▼

Photo:



Course Details:

Title:

Bsc in IT

Duration:

4

▼

Faculty:

Bsc in IT

BACK TO MAIN

Form 1